Project Final Report

PROCTECH 2SE3 – Systems Engineering Fundamentals

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Abstract

This report is to detail the development of an automated Coin Sorting and Counting System designed to solve inefficiencies and human errors associated with manual coin handling in banking and retail environments. The system uses an automatic feeder, conveyor belt transport, and high-speed IR sensors to identify, sort, and count standard Canadian coinage **(5¢, 10¢, 25¢, $1, and $2)** which is based off **Design Alternative 4**: **Fast Processing.** This design enables key functional features which include real-time monetary calculation, foreign object rejection, and mechanical jam detection with automated recovery sequences.

Through systematic design selection and pairwise comparison, the system was optimized with a heavy weight on operational efficiency and accuracy. This is to achieve a processing rate of **at least 60 coins per minute** with a sorting **accuracy of 99.5%**.

The implementation of fail-safe mechanisms such as data retention during power loss and automated bin-capacity monitoring prevents storage overflows. The system provides a reliable, high-throughput solution for coin processing, successfully balancing high-speed performance with safety and precision standards required for commercial operation. Important achievements include incorporation of safety measures like preservation of information after power failures and automatic capacity detection of bins to prevent overflows. To conclude, the project developed an autonomous coin processing system with fast speed and proper safeguards required by commercial applications.

Table of Contents

1. Project Description

The Coin Sorting and Counting System is designed to sort, count, and identify different coins autonomously. The system is intended for use in banks and retail environments, where it improves operational efficiency by reducing manual handling and minimizing human error.

The primary objective of the system is to ensure accurate and reliable coin processing under varying operating conditions. It maintains consistent performance by continuously monitoring system operation and ensuring stable processing of coins in real time. The system is designed to operate efficiently while maintaining safety and reliability during both normal operation and unexpected conditions.

In addition, the system incorporates automated monitoring and self-checking capabilities to detect irregularities, support maintenance, and ensure long-term operational stability. These features allow the system to maintain performance while reducing downtime and manual intervention.

By automating the coin sorting and counting process, the system enables users to quickly deposit mixed coins and receive an accurate digital summary of total monetary value, improving both efficiency and user experience.

2. Concepts of Operation (ConOps)

The following concepts of operation describe how the automated coin sorting and counting system operates under different conditions.

ConOps 1: User Interaction and System Initialization
The user loads mixed coins into the input hopper and starts the system using control buttons. The system initializes, performs basic checks, and provides status feedback through a display interface.

ConOps 2: Continuous Sorting and Counting Operation

The system processes coins continuously, identifying denominations and routing them into appropriate bins. During operation, it updates the total count and monetary value in real time while maintaining a steady processing rate.

ConOps 3: Coin Validation and Rejection

The system evaluates each input item to determine whether it is a valid coin. Invalid objects such as foreign coins or debris are automatically separated and redirected to a reject tray.

ConOps 4: Batch Completion and System Reset

When no additional coins are detected, the system completes the operation and generates a summary of total value and coin counts. The system then resets internal counters and returns to an idle state for the next user.

ConOps 5: Fault Detection and Recovery (Jamming)

The system continuously monitors coin flow to detect mechanical blockages. Upon detecting a jam, the system halts operation to prevent damage, attempts automatic recovery, and alerts the user if manual intervention is required.

ConOps 6: Capacity Monitoring and Overflow Prevention

The system tracks the fill level of each storage bin. When a bin approaches capacity, the system alerts the user and prevents overflow by stopping intake or redirecting coins.

ConOps 7: Maintenance and System Diagnostics

The system performs periodic self-checks to monitor internal component health and performance. It tracks usage and notifies the user when maintenance is required or when abnormal behavior is detected.

3. Objectives

3.1 Categorization

The objectives of the automated coin sorting and counting system are categorized into the following major groups:

Functional Performance

1. The system should accept a mixed input batch of standard Canadian coinage (5¢, 10¢, 25¢, $1, $2) without manual pre-sorting.

2. The system should accurately sort Canadian coin denominations.

3. The system should process coins with high efficiency.

4. The system should calculate the total monetary value of the processed batch in real time.

5. The system should maintain an individual count for each coin denomination.

6. The system should prevent foreign objects from entering valid coin storage.

System Reliability & Safety

7. The system should detect mechanical jams promptly.

8. The system should automatically halt operation upon jam detection.

9. The system should attempt an automated jam clearance sequence.

10. The system should retain data during power loss.

11. The system should operate at a safe noise level.

User Interface & Operational Feedback

12. The system should provide a session reset function.

13. The system should display total value and coin breakdown.

14. The system should detect idle state and stop operation automatically.

15. The system should display maintenance alerts.

16. The system should allow manual pause through an emergency stop.

Capacity & Maintenance

17. The system should monitor bin fill levels.

18. The system should alert when bins approach capacity.

19. The system should redirect coins or stop intake when bins are full.

20. The system should track total usage for maintenance scheduling.

21. The system should allow easy removal of storage containers.

22. The system should perform a self-check on startup.

3.2 Pairwise Comparison Chart

Objectives	Accuracy	Safety	Reliability	Operational Efficiency	SCORE
Accuracy	***	1	1	1	3
Safety	0	***	1	1	2
Reliability	0	0	***	1	1
Operational Efficiency	0	0	0	***	1

Accuracy

How accurately the system can perform its functions. Accuracy was given a score of 3 as the primary function of the system is to be able to correctly identify, count, and sort Canadian coins. Incorrect sorting or counting directly defeats the purpose of the system, despite speed or capacity.

Safety

The safety of the system for any users and operators. Safety was given a score of 2 since the protection of the operator and system must be ensured. Features such as fault detection, emergency stops, and enclosed components are essential for safe operation in commercial environments.

Reliability

Reliability was given a score of 1, as for long-term use, consistent performance is critical. The system must be able to preserve data during power loss, detect jams, and operate without frequent failure. If the system is reliable, that minimizes downtime and maintenance requirements.

Operational Efficiency

How efficiently the system executes its functions. Operational Efficiency was given a score of 1, as it must not come at the expense of accuracy, safety, or reliability. It can only be met after all other core objectives are satisfied.

4. Requirements

4.1 Requirements in Categories and Verification method

Functional

1. The system shall accept mixed standard Canadian coin denominations (5¢, 10¢, 25¢, $1, $2) without manual pre-sorting.

Verification: Test – Insert mixed coins and confirm acceptance without pre-sorting.

2. The system shall correctly identify and sort valid Canadian coins into their respective denomination bins.

Verification: Test – Compare sorted output with expected results using a known dataset.

3. The system shall physically reject non-coin objects and invalid items into a dedicated reject tray.

Verification: Test – Insert foreign objects and verify they are directed to the reject tray.

4. The system shall calculate and store the total monetary value of coins processed during each operating session.

Verification: Analysis – Compare calculated value with manual calculation.

Performance

5. The system shall process coins at a minimum rate of 60 coins per minute.

Verification: Test – Measure number of coins processed per minute.

6. The system shall achieve a sorting accuracy of at least 99.5%.

Verification: Test – Compare sorted results with known input values.

7. The system shall detect and reject non-coin objects with at least 95% reliability.

Verification: Test – Introduce invalid objects and evaluate rejection rate.

8. The system shall complete the identification, sorting, and counting of each coin within 0.5 seconds.

Verification: Test – Measure processing time per coin.

9. The system shall operate continuously for a minimum of 8 hours without failure.

Verification: Test – Run system continuously and monitor performance.

Constraint

10. The system shall stop accepting coins within 0.5 seconds of a mechanical jam.

Verification: Test – Introduce blockage and measure response time.

11. The system shall preserve coin count and total value during power loss.

Verification: Test – Interrupt power and verify data retention.

12. The system shall prevent operator access to internal components during operation.

Verification: Inspection – Check enclosure prevents access.

13. The system shall include an emergency stop mechanism within arm's reach.

Verification: Demonstration – Activate emergency stop and observe system response.

14. The system shall operate using standard 120 V AC power supply.

Verification: Inspection – Verify power input specifications.

Interface

15. The system shall provide a coin input hopper interface.

Verification: Inspection – Confirm presence of input hopper.

16. The system shall provide separate output bins for each denomination.

Verification: Inspection – Confirm physical output bins.

17. The system shall provide a reject tray for invalid objects.

Verification: Inspection – Confirm reject tray exists and functions.

18. The system shall provide a user interface displaying total value and coin counts.

Verification: Demonstration – Observe display output during operation.

19. The system shall provide visual status indicators.

Verification: Demonstration – Verify LED/status indicators during operation.

Safety

20. The system shall provide visual and audible alarms for faults or jams.

Verification: Demonstration – Trigger fault condition and observe alerts.

21. The system shall ensure all moving components are enclosed.

Verification: Inspection – Verify protective enclosure.

22. The system shall shut down if access panel is opened.

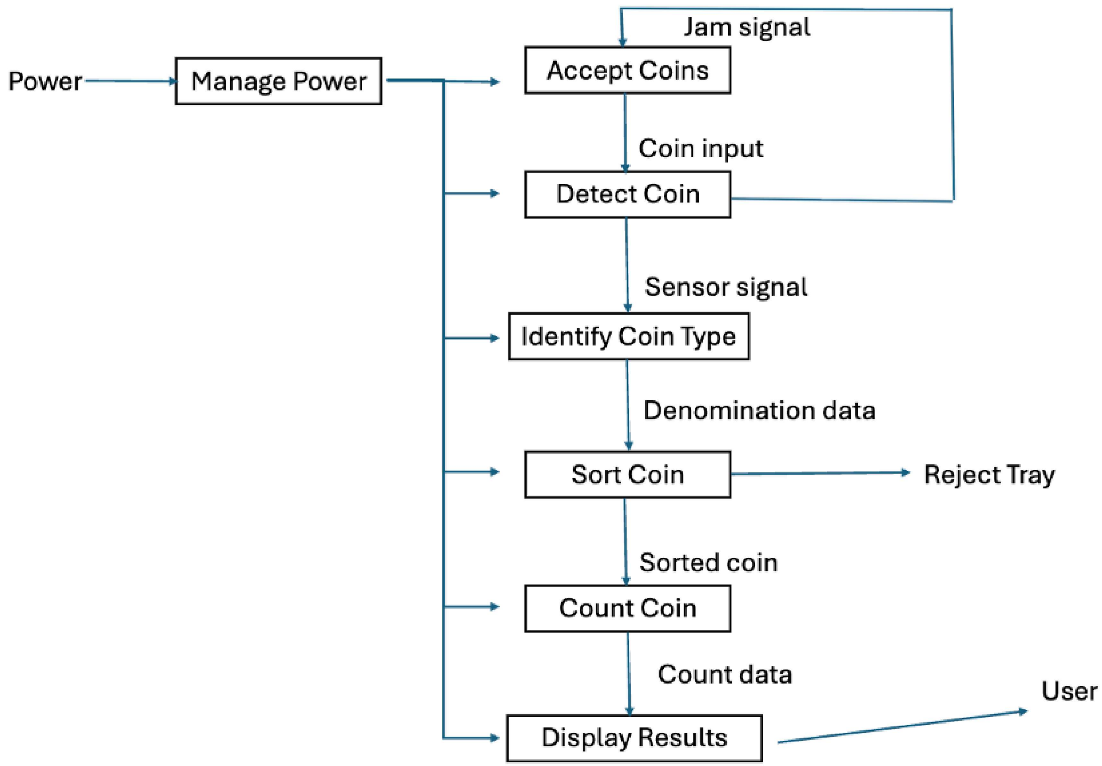
Verification: Demonstration – Open panel during operation and observe shutdown.

23. The system shall prevent operation without safety guards.

Verification: Demonstration – Attempt operation without guards installed.

5. Design

5.1 Functional Design



5.2 Morphological Chart

Function	Option 1	Option 2	Option 3
Coin Input Method	Coins are inserted manually into the machine.	Coins are inserted automatically using a feeder.	Coins are inserted through a guided slot.
Coin Detection	The system detects coins using an optical sensor.	The system detects coins using an IR sensor.	The system detects coins using a camera system.

Coin Identification	The system identifies coins by size.	The system identifies coins using image recognition.	The system identifies coins by weight.
Coin Transport	Coins move through the system using gravity.	Coins move using a rotating disk.	Coins move using a conveyor belt system.
Coin Sorting	Coins are separated using a rotating sorter.	Coins are separated using mechanical gates.	Coins are separated using motorized gates.
Coin Counting	Coins are counted using encoder.	Coins are counted using pulse signals.	Coins are counted using optical sensors
Reject Handling	Invalid coins are returned to the user.	Invalid coins go to a reject tray.	Invalid coins are stored in a reject container.
Result Display	Results are displayed using an LED display.	Results are displayed using an LCD screen.	Results are displayed using a touchscreen display.

5.3 Design Alternatives

Alternative 1: Ease of Use

This design focuses on minimizing user effort by using an automatic feeder and a touchscreen display for intuitive interaction. Invalid coins are returned directly to the user to reduce inconvenience. Compared to other alternatives, this option prioritizes user experience and ease of operation, but does not focus on maximizing speed or minimizing cost.

Alternative 2: Lowest Cost

This design minimizes manufacturing cost by using manual coin input, gravity-based transport, and basic optical sensors for detection. Sorting is achieved using simple mechanical gates. Compared to other alternatives, this option avoids expensive components such as motors and camera systems, resulting in lower cost but reduced performance and automation.

Alternative 3: Highest Accuracy

This design prioritizes precision by using a guided slot for consistent coin alignment and a camera system for image-based identification. Sorting is performed using motorized gates and counting is tracked using an encoder. Compared to other alternatives, this option achieves the highest accuracy but introduces greater system complexity and cost.

Alternative 4: Fast Processing

This design is optimized for high throughput by using an automatic feeder, conveyor belt transport, and fast IR sensors for detection. Coins are sorted using a continuous rotating mechanism. Compared to other alternatives, this option maximizes processing speed but may sacrifice some accuracy and increase system complexity.

Alternative Mapping

The following mapping shows how each design alternative selects specific options from the morphological chart.

Alternative 1: Ease of Use

- Input: Automatic feeder
- Detection: Optical sensor

- Identification: Size-based
- Transport: Gravity
- Sorting: Mechanical gates
- Counting: Pulse signals
- Reject Handling: Returned to user
- Display: Touchscreen

Alternative 2: Lowest Cost

- Input: Manual insertion
- Detection: Optical sensor
- Identification: Size-based
- Transport: Gravity
- Sorting: Mechanical gates
- Counting: Pulse signals
- Reject Handling: Reject tray
- Display: LED display

Alternative 3: Highest Accuracy

- Input: Guided slot
- Detection: Camera system
- Identification: Image recognition
- Transport: Rotating disk
- Sorting: Motorized gates
- Counting: Encoder
- Reject Handling: Reject tray
- Display: LCD display

Alternative 4: Fast Processing

- Input: Automatic feeder

- Detection: IR sensor
- Identification: Size-based
- Transport: Conveyor belt
- Sorting: Rotating sorter
- Counting: Optical sensors
- Reject Handling: Reject tray
- Display: LCD display

5.4 Design Selection

Objectives	Accuracy	Safety	Reliability	Operational Efficiency	SCORE
Accuracy	***	1	1	1	3
Safety	0	***	1	1	2
Reliability	0	0	***	1	1
Operational Efficiency	0	0	0	***	1

Alternative	Accuracy (0.43)	Safety (0.29)	Reliability (0.14)	Efficiency (0.14)	Total Score
Ease of Use	6	7	6	8	6.57
Lowest Cost	5	6	5	5	5.29
Highest Accuracy	10	8	7	7	8.71
Fast Processing	8	7	7	10	8.00

- Accuracy (Weight 0.43): It is assigned the highest weight because the primary function of the system is to correctly identify, count, and sort coins. When comparing Accuracy vs. Operational Efficiency, Accuracy wins, as a fast but incorrect system fails its purpose. When comparing Accuracy vs. Reliability, ensuring correct results is more important than consistent but inaccurate operation.

- Safety (Weight 0.29): Safety is given high importance to ensure protection of users during operation. The system must include features such as emergency stops, enclosed moving components, and fault detection to prevent accidents in a commercial environment.

- Reliability (Weight 0.14): Reliability is important for consistent long-term performance. The system must operate continuously, detect jams, and retain data during power interruptions. While important, it is ranked below accuracy and safety.

- Operational Efficiency (Weight 0.14): Operational efficiency is assigned the lowest weight because it should not compromise accuracy, safety, or reliability. While fast processing is beneficial, it is only valuable if the system maintains correct and safe operation.

Justification of Design Alternative Scores

Alternative 1: Ease of Use

- Accuracy (6): Uses basic sensing methods with moderate accuracy.
- Safety (7): Includes standard safety features such as enclosure and basic fault handling.
- Reliability (6): Moderate reliability due to dependence on user interaction and simpler components.

- Operational Efficiency (8): Automatic feeder improves processing flow and reduces user effort.

Alternative 2: Lowest Cost

- Accuracy (5): Limited accuracy due to simple sensing methods.
- Safety (6): Basic safety features are present but not highly advanced.
- Reliability (5): Lower reliability due to minimal system complexity and lack of automation.
- Operational Efficiency (5): Slower performance due to manual input and passive transport mechanisms.

Alternative 3: Highest Accuracy

- Accuracy (10): Image recognition system provides the highest level of accuracy.
- Safety (8): Well-controlled system with proper enclosure and safety measures.
- Reliability (7): Reliable but dependent on more complex components.
- Operational Efficiency (7): Slightly reduced efficiency due to processing time required for image analysis.

Alternative 4: Fast Processing (Selected Design)

- Accuracy (8): Provides high accuracy using IR sensors, though slightly lower than camera-based systems.
- Safety (8): Includes proper safety mechanisms and enclosed components.
- Reliability (7): Good reliability with automated processes and reduced manual interaction.

- Operational Efficiency (10): Optimized for high throughput and continuous operation.

5.5 Selected Design Statement

The selected design for the automated coin counter is Alternative 4: Fast Processing. This option focuses on handling coins quickly while still maintaining good accuracy and reliable operation.

In this design, coins are fed into the system using an automatic feeder, which keeps a steady flow of coins moving through the machine. A conveyor belt is used to transport the coins smoothly. For detection, IR sensors are used because they can identify coins quickly without the delay of more complex systems like cameras. The coins are then sorted using a rotating mechanical sorting mechanism, which directs them into the correct bins. Any invalid coins are sent to a reject tray, and the results are shown to the user through an LCD display.

Overall, this design includes all the main elements from the morphological chart, such as feeding, transport, detection, sorting, and display. It provides a good balance between speed, accuracy, and reliability, making it suitable for environments where a large number of coins need to be processed efficiently.

6. Interfaces

The external interfaces of the Automated Coin Sorting and Counting System are categorized into four categories: physical, mass, energy, and information interfaces. Each interface is clearly described along with its purpose.

6.1 Physical Interfaces

1. Coin Input Hopper

Allows the user to input mixed Canadian coins into the system. It serves as the primary entry point for all coins.

2. Output Collection Bins

Stores sorted coins by denomination (nickel, dime, quarter, loonie, toonie) and allows users to retrieve them after processing.

3. Reject Tray

Collects invalid items such as foreign coins, damaged coins, or non-coin objects to prevent contamination of valid outputs.

4. User Control Buttons (Start/Stop/Emergency Stop)

Allows the operator to start, stop, or immediately stop the system for safe operation.

5. Machine Housing and Enclosure

Protects internal components and prevents users from coming into contact with moving parts during operation.

6. Access Panel and Maintenance Cover

Provides access for maintenance, inspection, and cleaning of internal components.

6.2 Mass Interfaces

1. Mixed Coin Input Flow
 Represents the flow of mixed coins entering the system through the input hopper.

2. Sorted Coin Outputs
 Represents the separated coin streams exiting the system into individual bins for each denomination.

3. Reject Coin Flow
 Represents the flow of invalid or rejected coins directed into the reject tray.

4. Overflow Coin Flow
 Represents coins that are redirected when a bin reaches full capacity

6.3 Energy Interfaces

1. AC Power Input (120V)
 Provides electrical power from an external source required for system operation.

2. Internal Power Supply Interface
 Converts and distributes electrical energy to subsystems such as sensors, controller, and actuators.

3. Motor Power Interface
 Supplies electrical energy to motors responsible for coin transport and sorting mechanisms.

4. Sensor Power Interface
 Provides electrical energy to detection sensors used for coin identification and counting.

6.4 Information Interfaces

1. Sensor to Controller Signals

Transfers detection data such as coin presence, size and type from sensors to the control system.

2. Controller to Actuator Signals

Sends control commands to motors and sorting mechanisms to ensure coins are routed correctly.

3. Count Display Interface

Displays the total value and the count for each denomination to the user after processing.

4. System Status Indicators (LED and Display)

Communicates system states such as ready, running, jam, fault, and maintenance required.

5. User Input Commands

Transfers user commands such as start, stop, reset into the control system.

6. Fault and Jam Notification Interface

Provides alerts to the user when abnormal conditions such as jams or system failures occur.

7. Verification and Validation Plan

The following verification and validation processes ensure that the Automated Coin Sorting and Counting System meets all functional, performance, and safety requirements.

1. Coin Sorting Accuracy

- Verification
- Verify the system correctly separates each coin into its proper denomination bin.
- To be performed by the engineering and testing team
- Method: A controlled batch of mixed coins with known quantities will be processed and the results will be compared to the expected sorting outcomes.
- Equipment needed: Mixed coin sets, Collection bins, Data sheet
- Performed during prototype testing
- Ensure that the system meets the requirements of sorting at least 99.5% of coins into correct bins.

2. Coin Counting Accuracy

- Verification
- Verify displayed coin count matches the actual number of coins sorted.
- To be performed by the testing team
- Method: The system output will be compared with manual counting of coins after each test run.
- Equipment needed: Coin batches, Calculator, Data sheet
- Performed during prototype testing
- Ensure the system displays values that match actual counts with at least 99.5% accuracy.

3. Processing Speed

- Validation
- Validate the system meets the required processing speed.
- To be performed by the testing team

- Method: The number of coins processed within a set time will be measured.
- Equipment needed: Stopwatch, Test coin batch
- Performed during performance evaluation
- Ensure the system processes a minimum of 60 coins per minute.

4. Foreign Object Rejection

- Verification
- Verify that non-valid items are properly rejected by the system.
- To be performed by engineering and testing team
- Method: Invalid objects such as washers and foreign coins will be introduced and their output paths will be observed.
- Equipment needed: Test objects, Prototype system
- Performed during functional testing
- Ensure the system rejects at least 95% of invalid objects correctly.

5. Jam Detection

- Verification
- Verify the system can identify and respond to coin jams.
- To be performed by engineering and testing team
- Method: Intentionally block coin path and observe system response.
- Equipment needed: Coins, Obstruction setup
- Performed during safety testing
- Ensure the system detects jam and stops operation within 0.5 seconds.

6. Jam Recovery Function

- Validation
- Validate the system can attempt to clear jams automatically.

- To be performed by testing team
- Method: After creating a jam, the system's recovery sequence will be observed.
- Equipment needed: Prototype system, Test coins
- Performed during functional testing
- Ensure the system attempts to clear the jam and either resumes operation or alerts user.

7. Overflow Detection

- Verification
- Verify the system detects full bins and redirects coins.
- To be performed by testing team
- Method: Fill bin to capacity and observe system behavior.
- Equipment needed: Coins, full bin setup
- Performed during capacity testing
- Ensure the system redirects coins or stops intake when bin is full.

8. Real Time Update Performance

- Validation
- Validate the system updated coin counts in real time during operation.
- To be performed by testing team
- Method: Observe display updates while coins are being processed and compare with actual coin flow.
- Equipment needed: Prototype system, Test coins
- Performed during integration testing
- Ensure the system displays updates correctly without delay and reflects the actual coin flow,

9. Power Failure Data Retention

- Verification
- Verify the system data is preserved during power interruptions.
- To be performed by engineering team
- Method: Power will be cut during operation and system recovery will be evaluated.
- Equipment needed: Power supply setup, Prototype system
- Performed during reliability testing
- Ensure coin counts and total values remain unchanged after restart.

10. User Operation Functionality

- Validation
- Validate user can operate system functions such as start, stop, reset correctly.
- To be performed by representative users
- Method: Users will operate the system without prior instructions while being observed.
- Equipment needed: Prototype system
- Performed during user testing phase
- Ensure users operate the system successfully without errors or assistance.